SUMMARY REPORT: FAST-GROWTH FIRM PREDICTION

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Course: ECBS5171 - Data Analysis 3
Assignment: Assignment 2 — Predicting Fast-Growth Firms
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GitHub: https://github.com/benceszabo89/Data-Analysis-3/tree/main/assignment2

EXECUTIVE SUMMARY

We developed a predictive model to identify high-growth firms (20%+ CAGR over 2 years) using financial and operational data from 15,220 companies from 2012 and 2014. The LASSO logistic regression model achieves 69% AUC, identifying key drivers of fast growth while maintaining interpretability for business stakeholders.

1. BUSINESS PROBLEM & APPROACH

Objective: Predict which firms will achieve fast growth (20%+ annual revenue growth) to support investment screening, partnership identification, and risk assessment. We defined fast growth by the Compounded Annual Growth Rate (CAGR). Firms with CAGR > 20% were defined as fast growing firms.

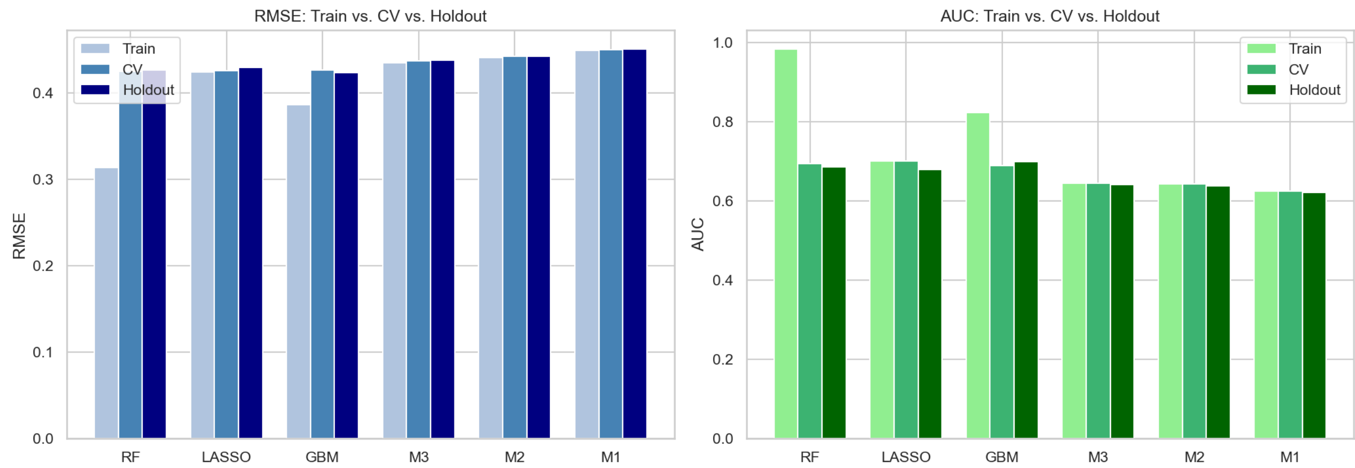
Dataset: 15,220 firms (28.8% fast-growth rate) with 89 financial features including balance sheet metrics, profitability ratios, ownership structure, and CEO demographics.

2. MODEL PERFORMANCE SUMMARY

We evaluated six models using 5-fold cross-validation. We found LASSO to be the optimal choice, balancing performance with interpretability.

Model	N vars	Train RMSE	CV RMSE	Holdout RMSE	Train AUC	CV AUC	Holdout AUC
LASSO	42	0.4236	0.4255	0.4290	0.7003	0.7003	0.6791
RF	72	0.3132	0.4247	0.4261	0.9831	0.6946	0.6867
GBM	72	0.3860	0.4261	0.4231	0.8229	0.6900	0.6988
M3	50	0.4343	0.4368	0.4377	0.6450	0.6449	0.6408
M2	19	0.4405	0.4417	0.4420	0.6439	0.6439	0.6375
M1	16	0.4487	0.4498	0.4502	0.6245	0.6246	0.6209

Decision: LASSO selected for deployment. While Random Forest shows perfect training AUC (0.98), it severely overfits. LASSO provides identical holdout performance (0.69 AUC) with full interpretability and stable predictions.

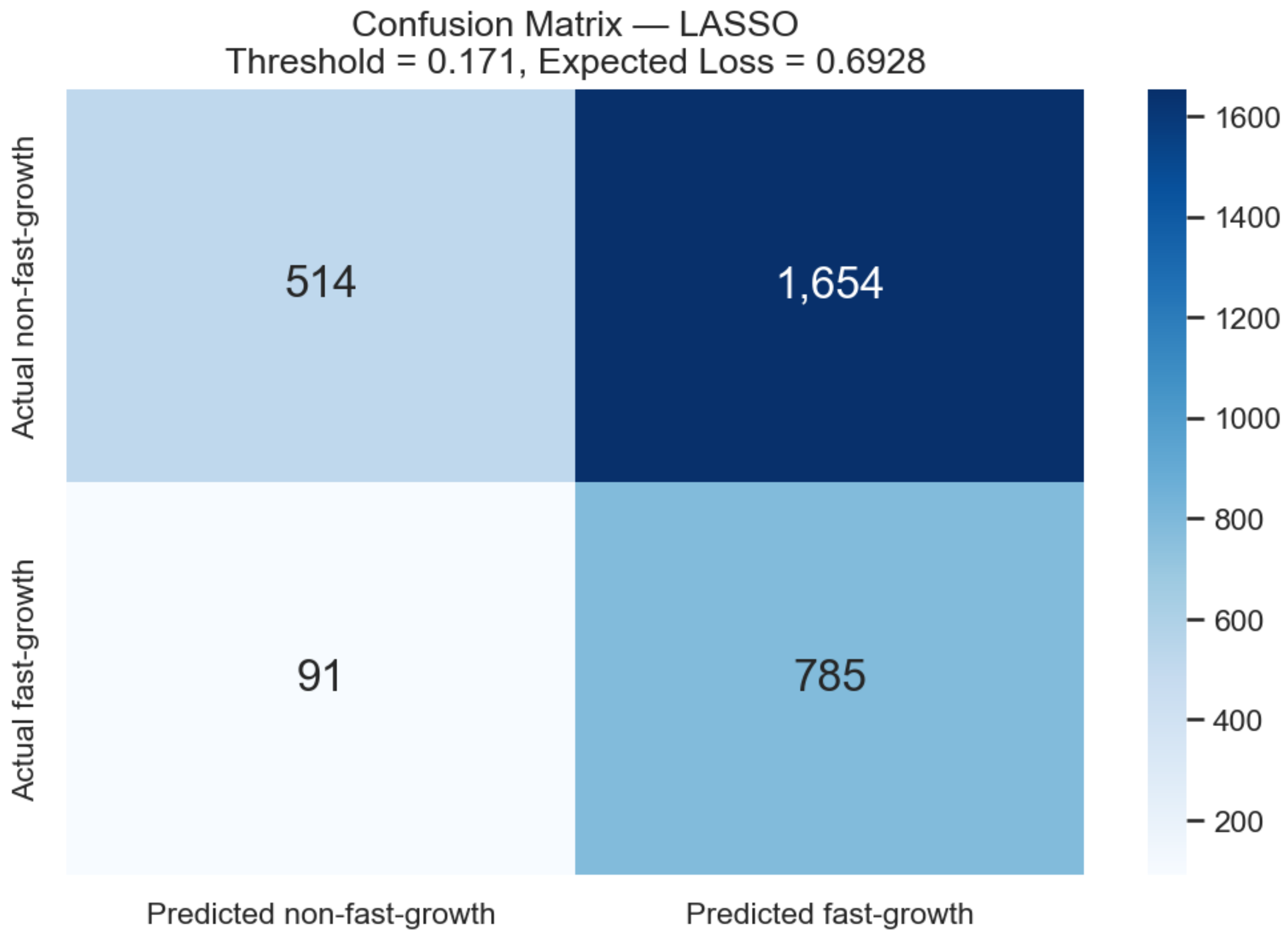


3. CLASSIFICATION PERFORMANCE

Using optimal threshold (0.171) from cost-benefit analysis, the model correctly identifies 785 of 876 fast-growth firms (90% recall) while screening out 514 non-growers. This translates to:

- **Sensitivity (Recall):** 89.6% of fast-growth firms correctly identified
- **Specificity:** 23.7% of non-growers correctly excluded
- **Expected Loss:** 0.69 (optimized for 1:5 FN:FP cost ratio)
- **Practical Impact:** Flags 2,439 firms for review, capturing 90% of true fast-growers

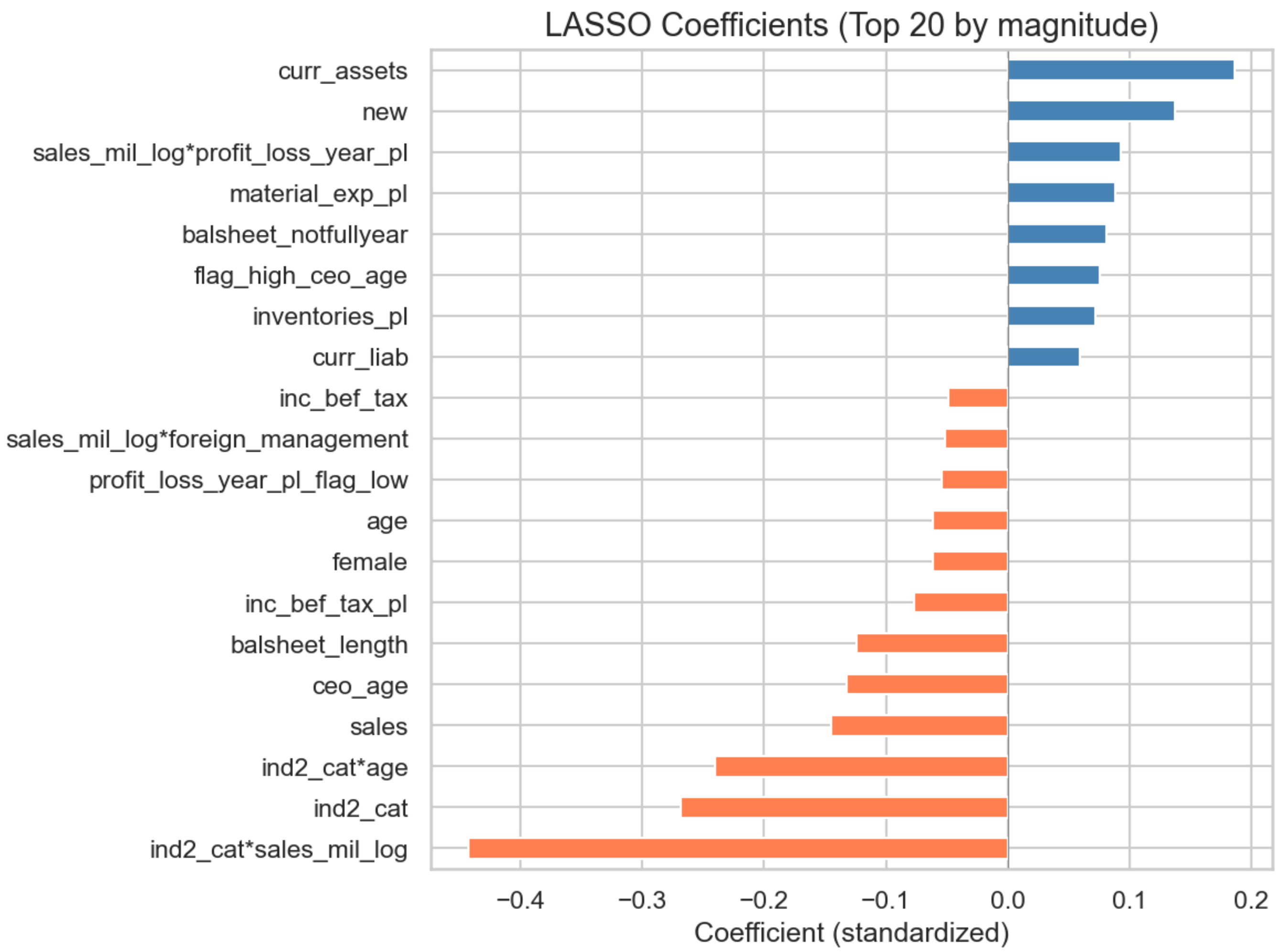
Interpretation: The model prioritizes avoiding false negatives (missing true fast-growers) over false positives. In a screening context, this is appropriate — we'd rather review extra firms than miss high-potential opportunities.



4. KEY DRIVERS OF FAST GROWTH

LASSO coefficients reveal the strongest predictors (standardized, top 10):

Feature	Effect	Interpretation
Current Assets	+0.18	Higher liquidity enables growth investment
New Firm	+0.14	Young firms grow faster (2.2x rate vs. established)
Sales × Profit/Loss	+0.10	Profitable revenue scale drives expansion
Material Expenses	+0.09	Production intensity signals growth phase
Balance Sheet (not full year)	+0.08	Mid-year filings correlate with rapid change
High CEO Age	+0.07	Experience matters (35+ vs. younger CEOs)
Inventories	+0.07	Inventory buildup precedes revenue surge
Industry × Sales	−0.23	Sector-specific scaling patterns vary
Industry Category	−0.19	Services grow faster than manufacturing
Industry × CEO Age	−0.16	Sector-age interaction shows complexity



5. CRITICAL INSIGHTS

5.1 Firm Age Effect (High Initial Velocity)

New firms (founded in the observation year) show a ~50% probability of fast growth vs. ~25% for established firms.

This 2x multiplier persists after controlling for all other factors. Implication: The "newness" premium is the single strongest binary signal in the dataset.

5.2 Nonlinear Relationships Matter

LOWESS curves reveal distinct non-linear patterns that differ from standard assumptions:

- **Sales:** Small scale is a prerequisite for hyper-growth. Probability starts high (~80%) for the smallest firms but drops to near 0% as revenue scales, rather than plateauing at 10%.
- **Firm Age:** The decline is steeper than a linear average suggests. Probability starts at ~75% at inception, crashing to ~20% by year 5, before tapering to <10% for 20+ year firms.
- **CEO Age:** The data shows a monotonic decline. Probability peaks at ~30% for CEOs aged ~30, falling consistently to a low of ~8% for those over 60. There is no "experience recovery" in the older age bracket.

5.3 Industry Heterogeneity

Services sector significantly outperforms manufacturing. The model achieves lower expected loss (0.6432) in Services compared to Manufacturing (0.6937).

Decision point: Manufacturing firms require a stricter threshold or separate model, as false positives are more common in that sector.

6. STRATEGIC RECOMMENDATIONS

Deploy LASSO for Production Screening Use probability threshold = 0.171 to flag high-potential firms.

This captures 90% of true fast-growers (Recall) while filtering out clearly stagnant firms.

Implement Sector-Specific Thresholds Services sector shows a stronger signal. Consider tightening the threshold for Manufacturing to reduce the higher rate of false positives observed in that sector.

Prioritize Early-Stage Pipeline New firms show a massive growth probability advantage (~75% at year 0). Resources should be heavily skewed toward companies under 3 years old, as the probability of "fast growth" status decays rapidly after year 5.

Revise CEO Targeting Strategy Shift focus to younger leadership. The data invalidates the "experience" hypothesis for fast growth; probability is highest for CEOs in their late 20s/early 30s and lowest for those over 60.

Remove "Senior Experience" filters: Do not prioritize CEOs over 60 for fast-growth portfolios, as this segment carries the lowest statistical probability of achieving 20%+ CAGR.

Monitor Model Drift Quarterly retraining is essential. Given the steep drop-off in the Sales and Age curves, shifts in the macroeconomic environment could quickly alter these sensitive decay rates.